

VISVESVARAYA TECHNOLOGICAL UNIVERSITY
JNANA SANGAMA, BELAGAVI – 590018



A Synopsis on

**“Agentic AI to Augment Organization Specific Knowledge for
Real Time Response of LLM”**

Submitted for the Major Project Work-Phase 1
(Academic Year 2025–26)

Submitted by

P. Parker Vijay Harshan	1NT23IS149
Palak Pandit	1NT23IS150
P Charan Prashanth	1NT23IS148
Abhay Rawat	1NT23IS004

Under the Guidance of

Dr. Sudhir Shenai

Professor

Dept. of Information Science and Engineering



Department of Information Science and Engineering

Certificate

This is to certify that the **Phase-1** project work entitled “*Agentic AI to Augment Organization Specific Knowledge for Real Time Response of LLM*” has been carried out by **P. Parker Vijay Harshan (INT23IS149)**, **Palak Pandit (INT23IS150)**, **P Charan Prashanth (INT23IS148)**, and **Abhay Rawat (INT23IS004)**, bonafide students of *Nitte Meenakshi Institute of Technology*, in partial fulfillment of the requirements for the award of the degree of **Bachelor of Engineering** in the **Department of Information Science and Engineering**, under **Visvesvaraya Technological University**, Belagavi, during the academic year **2025–2026**.

The project report has been examined and approved as it meets the academic requirements specified under the autonomous scheme of **Nitte Meenakshi Institute of Technology** for the said degree.

Signature of the Guide

Dr. Sudhir Shenai

Professor
Nitte Meenakshi Institute of
Technology,
Bengaluru-560064

Signature of the HoD

Dr. Manoj Kumar MV

HoD, dept. of ISE
Nitte Meenakshi Institute of
Technology,
Bengaluru-560064

Signature of the Principal

Dr. H C. Nagaraj

Principal
Nitte Meenakshi Institute of
Technology,
Bengaluru-560064

External Viva-Voce

Name of Examiners

Signature with Date

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P. Parker Vijay Harshan	(1NT23IS149)
Palak Pandit	(1NT23IS150)
P Charan Prashanth	(1NT23IS148)
Abhay Rawat	(1NT23IS004)

Place: Bengaluru

Academic Year: 2025–2026

Academic Year: 2025–2026
Declaration by Students

We, the undersigned students **P. Parker Vijay Harshan (1NT23IS149)**, **Palak Pandit (1NT23IS150)**, **P Charan Prashanth (1NT23IS148)** and **Abhay Rawat (1NT23IS004)** hereby declare that we are fully aware of the **project conduction procedures** and **evaluation rubrics** as prescribed by the department. We understand that the **project work will be evaluated through continuous assessment** across the defined phases.

<i>Name of the Student</i>	<i>Signature</i>
P. Parker Vijay Harshan	
Palak Pandit	
P Charan Prashanth	
Abhay Rawat	

Confirmation by Guide

I, **Dr. Sudhir Shenai**, hereby declare that I shall guide the above-mentioned students in the project entitled “**Agentic AI to Augment Organization Specific Knowledge for Real Time Response of LLM**” and will ensure that the project work is carried out under my supervision and submitted as per the academic requirements for the academic year **2025–2026**.

Signature of the Guide

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1. Introduction

The primary objective of the proposed project is to design and develop an AI-powered Moodle Assistant capable of simplifying academic knowledge retrieval through a conversational interface. The system is intended to reduce the dependency on manual navigation across multiple Moodle dashboards and provide users with faster and more intuitive access to institutional information. The project focuses on demonstrating the feasibility of combining artificial intelligence, database retrieval, and role-based access control into a unified academic assistant platform [7, 8, 9, 15].

One of the key objectives of the system is to provide real-time, context-aware assistance for students, faculty members, and administrators. Instead of manually searching through attendance records, grade reports, mentor details, or course information, users can directly ask questions using natural language. The assistant interprets the query, retrieves the required data from the Moodle database, and presents the information in a structured and user-friendly manner [2, 4].

Another important objective is to automate academic information access and reporting. The system aims to reduce repetitive manual work by enabling users to generate summaries, reports, and downloadable documents automatically. The project supports multiple output formats such as TXT, PDF, Word, and Excel so that retrieved information can be easily stored, shared, or submitted for academic purposes.

The project also aims to demonstrate the use of role-based access and secure information retrieval within an educational environment. Since different users require access to different categories of academic information, the assistant applies role-aware restrictions to ensure that only authorized information is displayed. This improves security, privacy, and institutional compliance while maintaining usability [15].

A further objective of the project is to create a modular and scalable architecture that can be extended in future phases. The current prototype focuses on demonstrating the core workflow of role detection, query classification, SQL-based retrieval, and AI-generated responses. However, the architecture is designed in such a way that future enhancements such as semantic search, advanced intent classification, recommendation systems, predictive analytics, and integration with additional institutional services can be implemented without redesigning the complete application [8, 7].

Students can quickly retrieve their records, faculty members can access course-related information, and administrators can simplify routine academic interactions. By integrating AI technologies into academic systems, the project demonstrates a modern approach toward intelligent educational assistance and automation.

2. Objectives

Provide Real-Time Academic Assistance. The system is designed to respond to academic queries instantly through natural language interaction so that users do not need to manually navigate across multiple Moodle portals.

Enable Context-Aware Information Retrieval. The assistant is intended to understand user roles and query intent so that it can retrieve relevant academic details such as attendance, grades, mentor information, and enrolled courses accurately.

Automate Academic Data Access and Reporting. The project aims to reduce repetitive manual work by enabling users to generate summaries and retrieve institutional information automatically without requiring technical knowledge of database systems or portal navigation.

Support Multiple Output Formats. The system provides the ability to export generated responses and reports in formats such as TXT, PDF, Word, and Excel for academic and administrative use.

Implement Role-Based Access Control. The assistant ensures secure and controlled access to institutional information by restricting data visibility according to user roles such as student, faculty member, and administrator.

Improve Productivity and User Experience. By reducing the time spent searching for academic records, the system improves communication efficiency through a unified conversational interface.

Demonstrate Integration of AI with Moodle. The prototype shows the feasibility of combining FastAPI, MySQL, Groq LLM services, and Moodle database systems to create an intelligent educational assistant.

Develop a Modular and Scalable Architecture. The system is built using modular components so that advanced features such as semantic search, predictive analytics, and intelligent recommendations can be added in future phases.

Enhance Accessibility of Academic Information. The project makes institutional data easier to access and understand for users with varying levels of technical expertise through natural language interaction.

3. Literature Survey and Review

Learning Management Systems such as Moodle, Blackboard, and Google Classroom are widely used in educational institutions for managing attendance, grades, assignments, examinations, course materials, and academic communication. These systems provide centralized access to academic information and simplify institutional management processes [11, 12, 13, 14].

Existing academic portals mainly rely on menu-driven interfaces where users must manually navigate through multiple dashboards, pages, and sections to retrieve specific information. This process often becomes repetitive, time-consuming, and inconvenient for students and faculty members [13, 11].

Rule-based academic chatbot systems have been introduced in some institutions to answer predefined queries related to schedules, assignments, examinations, and academic notifications. These systems improve accessibility but are limited in understanding flexible and context-aware user queries [3, 2].

Artificial Intelligence based virtual assistants use Natural Language Processing techniques to provide conversational interaction between users and systems. Such assistants improve user experience by allowing users to communicate using natural language instead of traditional search mechanisms [1, 4].

Retrieval-Augmented Generation systems combine structured database retrieval with AI-generated responses to improve the accuracy, reliability, and contextual relevance of generated information. These systems are increasingly used in domain-specific AI applications [7, 8, 9, 10].

Role-based information systems are commonly used in educational and enterprise environments to ensure secure and controlled access to information. Different users such as students, faculty members, and administrators are provided access according to their permissions and responsibilities [15].

Educational analytics and recommendation systems analyze academic performance data to generate insights related to attendance trends, student performance, and learning outcomes. These systems support decision-making and improve academic monitoring [13, 14, 12].

4. Limitations of the Existing Work (Gap Analysis) and Motivation

Complex Navigation in Existing Academic Systems. Most academic management systems require users to manually navigate through multiple dashboards, menus, and modules to retrieve information such as attendance, grades, assignments, and mentor details. This process becomes time-consuming and reduces user convenience [13, 11].

Lack of Conversational Interaction. Traditional academic portals are designed using menu-driven workflows rather than natural language interaction. Users must already know where information is stored within the system, making retrieval difficult for new or less experienced users [3, 2].

Fragmented Academic Information. Academic data is often distributed across different sections and modules of the portal. Users are forced to perform repetitive searches across attendance systems, grade sections, course pages, and communication portals to obtain complete information [13, 14].

Limitations of Rule-Based Chatbots. Existing rule-based academic chatbot systems can answer only predefined questions and lack the capability to understand flexible, complex, or context-aware user queries effectively [3, 2, 4].

Limited Database Integration in AI Assistants. Many AI-based virtual assistants are not directly connected to trusted institutional databases. As a result, they may generate incomplete, inaccurate, or unreliable academic information [7, 8, 10].

Insufficient Role-Based Personalization. Several academic systems do not dynamically adapt responses according to the role of the user such as student, faculty member, or administrator. This reduces both usability and security in academic environments [15].

Motivation for the Proposed System. The proposed Moodle AI Assistant is motivated by the need to develop a secure, conversational, role-aware, and AI-powered academic information retrieval system capable of simplifying access to institutional data while improving user experience, accessibility, and productivity.

5. Functional Requirements for Prototype

The prototype implements a limited but meaningful set of core use cases that demonstrate the assistant's value. The emphasis is on showing the system's ability to identify the user, retrieve the correct academic data, and present the response in a usable format.

The implemented requirements are intentionally scoped to the primary success path so that the prototype remains realistic within the semester timeline.

Req ID	Functional Requirement (SMART compliant)	Expected Outcome
FR1	The system shall allow a user to enter a valid user ID and load the correct role-based academic context.	The user sees the appropriate profile, permissions, and overview for student, faculty, or admin access.
FR2	The system shall answer a natural-language academic query by retrieving the matching Moodle data and presenting it in the requested format.	The user receives a relevant response about profile, courses, grades, attendance, backlogs, or mentor information.
FR3	The system shall allow the user to export the generated answer as text, PDF, Word, or Excel.	The response can be downloaded in the selected output format for offline use or submission.

Table 5.1 Functional Requirements Prototype

6. Proposed Methodology for Building Prototype

The prototype uses a retrieval-first, response-second workflow. The user sends a query through the web interface, the backend resolves the role, identifies the intent, fetches the relevant Moodle records, and then formats the response as structured text or a downloadable file [7, 8, 9, 15].

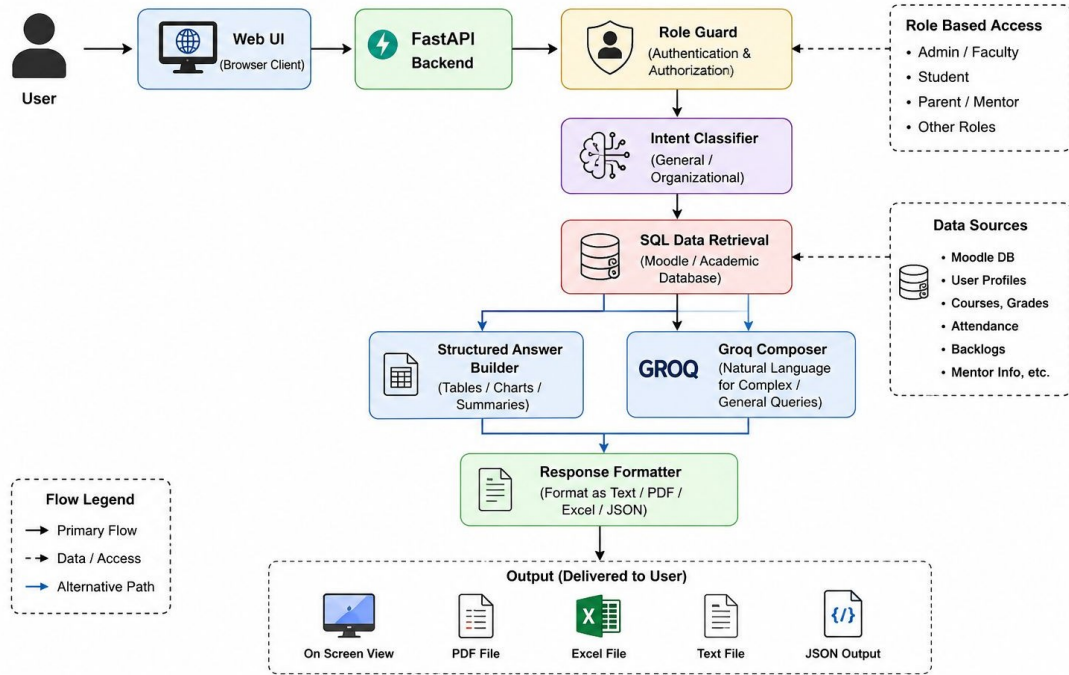


Figure 6.1 System Architecture of Moodle AI Assistant

6.1 Workflow / Process Steps

The workflow begins when a user enters a user ID and query through the browser interface. The system then detects the user's role, classifies the query as general or organizational in nature, retrieves the matching data from the Moodle SQL database, and generates a response that is formatted according to the requested output type. Finally, the output is displayed on screen or exported as a downloadable file.

6.2 Hardware Used

A standard laptop or desktop system for local deployment, a MySQL server or Docker-based MySQL container, and a browser client for the front-end.

6.3 Software and Programming Tools

Python, FastAPI, Uvicorn, PyMySQL, Groq API, HTML, CSS, JavaScript, openpyxl, python-docx, and fpdf2.

6.4 Algorithms or Techniques Implemented

Heuristic role detection, rule-based intent classification, SQL joins for academic data retrieval, and Groq-based natural-language composition for complex or general queries.

The design is intentionally modular so that the retrieval and response layers can be improved independently without rewriting the entire system.

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8. Annexures

8.1 Annexure-1: Prototype Description of Moodle AI Assistant

The proposed Moodle AI Assistant is developed as a **Prototype / Proof-of-Concept (PoC)** to demonstrate the feasibility of integrating Artificial Intelligence with academic management systems for intelligent knowledge retrieval [7, 8, 15].

The prototype focuses on validating the following core functionalities:

- **Conversational Academic Query Handling**

The system allows users to interact using natural language queries instead of manually navigating through multiple Moodle dashboards and menus.

- **Role-Based Information Retrieval**

The assistant identifies user roles such as student, faculty member, and administrator and retrieves information according to the permitted access scope.

- **SQL-Based Moodle Data Retrieval**

The prototype connects directly with Moodle database tables using MySQL and retrieves academic information such as attendance, grades, mentor details, and course enrollments.

- **AI-Powered Response Generation**

Groq LLM services are integrated to generate meaningful, structured, and context-aware responses for academic queries.

- **Multi-Format Report Generation**

The system supports exporting generated responses into multiple formats such as TXT, PDF, Word, and Excel for academic and administrative purposes.

- **Modular and Scalable Architecture**

The implementation separates workflow orchestration, database retrieval, response formatting, and access control into independent modules to support future scalability.

- **Prototype-Oriented Development**

The Phase-1 implementation focuses on demonstrating technical feasibility and core workflow validation rather than production-level optimization or deployment readiness.

8.2 Annexure-2: Functional Requirements of Moodle AI Assistant

Functional Requirements of the Proposed System

Functional requirements define the features and operations that the Moodle AI Assistant must perform to achieve the project objectives. These requirements describe the expected system behavior from the user's perspective and ensure that the prototype demonstrates the intended academic functionalities.

The functional requirements for the proposed system are designed to be **SMART compliant**, meaning they are Specific, Measurable, Achievable, Relevant, and Time-Bound within the scope of the prototype.

Each functional requirement follows the structure:

User Action → System Processing → Expected Response

The major functional requirements implemented in the Moodle AI Assistant prototype are listed below.

Req ID	Functional Requirement (SMART)	SMART Justification	Expected Outcome
FR1	The system shall allow users to enter a valid user ID and retrieve role-based academic information.	Specific: User identification; Measurable: Correct role detection; Achievable: Database lookup; Relevant: Core functionality; Time-Bound: Prototype implementation	The system identifies the user role and loads the corresponding academic information successfully.
FR2	The system shall process natural language academic queries and retrieve relevant Moodle data.	Specific: Query handling; Measurable: Response accuracy; Achievable: NLP and SQL integration; Relevant: Main objective; Time-Bound: Implemented in prototype	Users receive meaningful responses related to attendance, grades, courses, and mentor information.
FR3	The system shall generate downloadable academic reports in multiple formats.	Specific: File generation; Measurable: Successful export; Achievable: Existing Python libraries; Relevant: Reporting functionality; Time-Bound: Implemented in Phase-1	Users can export generated responses as TXT, PDF, Word, and Excel documents.

Table 8.1 SMART Functional Requirements of Moodle AI Assistant Prototype

Module ID	Module Name	Responsibility
M1	User Interface Module	Provides a browser-based conversational interface for user interaction and query submission.
M2	Role Detection Module	Identifies whether the user is a student, faculty member, or administrator based on institutional records.
M3	Intent Classification Module	Classifies user queries into academic, organizational, or general-purpose requests.
M4	Database Retrieval Module	Retrieves attendance, grades, courses, and mentor information from Moodle SQL tables.
M5	Response Generation Module	Uses Groq LLM services to generate conversational and structured responses.
M6	Report Export Module	Converts generated responses into TXT, PDF, Word, and Excel formats.

Table 8.2 Major Modules of Moodle AI Assistant System